Reg. No. | | | | | | | | | |

Question Paper Code : 18006

B.E / B.TECH. DEGREE EXAMINATION, NOVEMBER / DECEMBER 2021

Third Semester

B.E. – Computer Science and Engineering

19CS391 – PROGRAMMING WITH JAVA

(Regulations: Mepco – R2019)

Duration: 3 Hours Max. : 100 Marks

Answer ALL Questions

PART A – (10 × 2 = 20 Marks)

BTL, CO		
U,CO1	1.	What is the keyword used for declaring a constant variable? Justify your answer. A) const B) static C) final D) constant
U,CO1	2.	How do you convert an integer into a string?
A,CO2	3.	Which statements are correct regarding interfaces? Justify your answer. (i) Interface variables are by default public static and final (ii) An interface can implement another interface A) (i) is true and (ii) is false B) (ii) is true and (i) is false C) Both are false D) Both are true
U,CO2	4.	How do you invoke an overridden superclass method from the subclass?
U,CO3	5.	_____ is used to get the total no. of elements in an Array List. Justify your answer. A) length B) size C) array length D) capacity
U,CO3	6.	What is the purpose of generic class and generic method?
R,CO5	7.	Which of the following is passed as an argument to paint () method? Justify your answer. A) Image object B) Graphics object C) Canvas object D) Frame object
A,CO5	8.	Write a simple program that displays an empty frame on screen.
R,CO6	9.	Which of the following is a static member of the Thread class? Justify your answer. A) join B) run C) sleep D) start
U,CO6	10.	Why do we need join () method?

PART B – (5 × 16 = 80 Marks)

A,CO1 11. a) i. Write a method that computes the sum of the digits in an integer. Use the following method header.

public static int sumDigits(long n) (8 Marks)

U,CO1 11. a) ii. What is method overloading? Is it permissible to define 2 methods that have the same name but different parameter types? Is it permissible to define 2 methods in a class that have identical method names and parameter lists but different return value types or different modifiers? (8 Marks)

OR

A,CO1 11. b) i. Write a program that reads x and y coordinates of 2 points and displays the distance of their points. (8 Marks)

A,CO1 11. b) ii. Define a method named *smallest()* for finding the smallest of given n values. Write a program that reads in the temperature at a hill station on the 7 days of a week and then computes the minimum temperature for the week. (8 Marks)

A,CO2 12. a) i. Design an abstract class Shape. Inherit the abstract class Shape in Circle, Triangle and Square. Override the abstract method compute areal in all subclass. Write an application program to utilize the above classes. (8 Marks)

A,CO2 12. a) ii. Discuss is detail about functional interface with a suitable example. (8 Marks)

OR

A,CO2 12. b) i. Create a class Student with the data members Name, Register Number, Mark1, Mark2, Mark3, Total and Average. Using necessary member functions get the input, calculate Total and Average and display the student information's. If the register number exceeds 6 digits then raise one user defined exception called *InvalidRegNoException*. Similarly if marks value is greater than 100 throw an exception called *MarkOutOfBoundsException* (8 Marks)

A,CO2 12. b) ii. Discuss about the use of refection package with suitable example. (8 Marks)

A,CO3	13. a) i.	What is an ArrayList? Give an example. How will you remove duplicates from ArrayList?	(8 Marks)
U,CO3	13. a) ii.	How to make a Java class Serializable? Explain with an example.	(8 Marks)

OR

A,CO3	13. b) i.	Write a Java program to implement the generic stack.	(8 Marks)
U,CO4	13. b) ii.	How do you implement Java method in native code? Explain with an example.	(8 Marks)
U,CO5	14. a) i.	Describe Border Layout. How do you create a Border Layout manager? How do you add a component to a Border Layout container? Illustrate with an example.	(8 Marks)
A,CO5	14. a) ii.	Develop an applet that receives 2 whole numbers in two text-fields and displays the quotient and remainder in two other text-fields, when the button with the caption named "Divide" is clicked.	(8 Marks)

OR

U,CO5	14. b) i.	How do JDBC drivers help in database connectivity? Give an example program to demonstrate the same.	(8 Marks)
A,CO5	14. b) ii.	A car company called Yankee is selling six models of cars. Design a swing application with six JButtons along with the pictures of the cars. When a button is clicked, the details of the particular model must appear in an exclusive background color, textcolor and Font.	(8 Marks)
A,CO6	15. a) i.	Create 2 threads by implementing the runnable interface. First thread should print odd numbers and the second thread should print even numbers.	(8 Marks)
U,CO6	15. a) ii.	What is Thread synchronization? Explain with an example.	(8 Marks)

OR

A,CO6	15. b) i.	Define a thread using Thread class to generate the factorials of first 20 natural numbers. Create an instance for this thread and then activate it.	(8 Marks)
U,CO6	15. b) ii.	Define thread. Discuss in detail about thread states with a neat diagram.	(8 Marks)